

EVOLUTIONARY THINKING

WEEK 36 SESSION 1



DEPARTMENT OF MOLECULAR BIOLOGY AND GENETICS

AARHUS UNIVERSITY

EVOLUTIONARY THINKING
1. SEPTEMBER 2026

EMMA FEMKE DIEPEVEEN
PHD STUDENT



OVERVIEW OF TODAY

- Introduction of the TA
- Primate of the week
- Make pre-test
- Download MEGA
- Menti of things you would want more explanation about



EMMA

27 y.o.

Background:

- Bsc Medicine
- Bsc Biology
- Msc Evolution & Ecology

2nd year of the PhD

Comparative genomics
of primate sex
chromosomes



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Please e-mail me if you have questions besides TA session: emdi@birc.au.dk

My office is: 1872-354

Always allowed to bring cake

You will mainly see me on wednesdays because I have many courses/conferences on
fridays



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PRIMATE OF THE WEEK

Bonobo

Pan Paniscus

Hippie of the great apes

Sperm competition masters

Make love not war



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WHERE IS THE TA MATERIAL

The screenshot shows a Canvas LMS interface for a course titled "Evolutionary Thinking (E26.505212U001...)". The user is logged in as Emma Femke Diepeveen. The course navigation bar includes links for Course Home, Content, Activities, My Course, and Help. A banner indicates "Previewing Content as a Student".

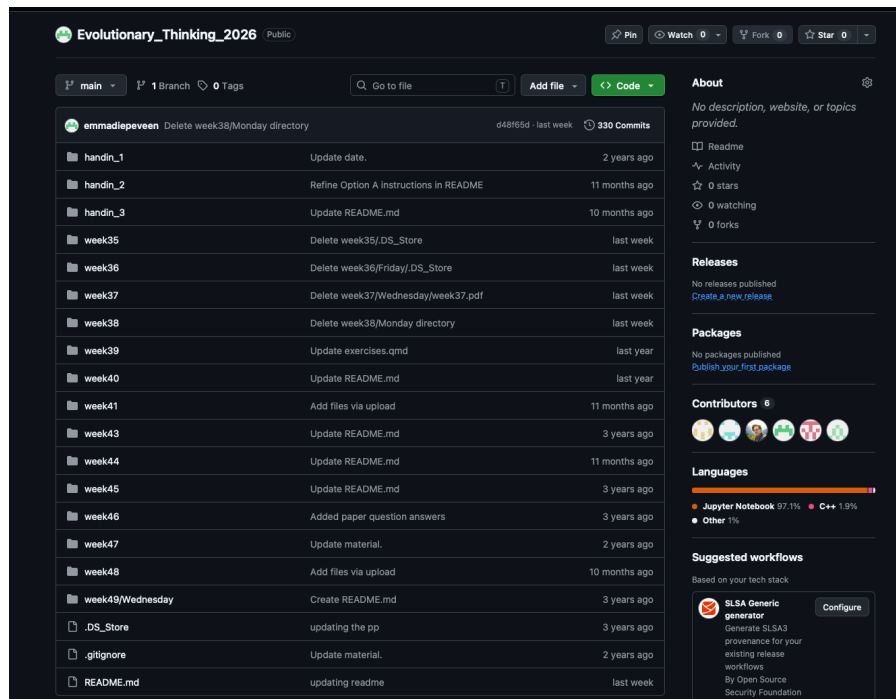
The left sidebar contains a "Weekly program" section with the following items:

- Link to TA session materials (0/19 Completed) - This item is circled in green.
- Week 35 - The origin of life and introduction to... (0/11 Completed)
- Week 36 - Molecular Phylogenetics (0/17 Completed)
- Course plan by week (Completed)
- Background literature (if needed) (0/2 Completed)
- Student presentations (0/7 Completed)
- Assignments
- Exam
- How to structure your course page
- Untitled
- Textbook chapters and papers for lectures and... (0/19 Completed)
- Student presentations (TRUE/FALSE questions)

The main content area shows the "Description" for the selected item: "Github with TA materials for every Wednesday and Friday". Below the description is a large blue link icon with an arrow pointing right, and the text "Open link in a new web browser tab." with an "Open Link" button.

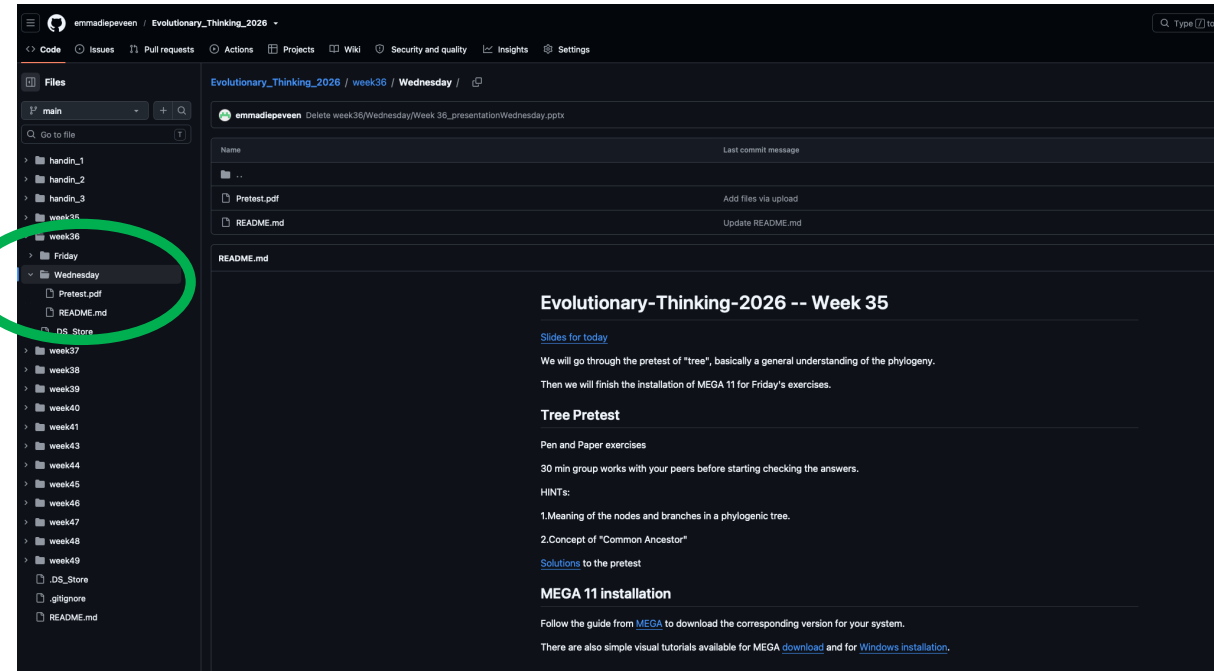


WHERE IS THE TA AVAILABLE



The screenshot shows the GitHub repository page for 'Evolutionary_Thinking_2026'. The left sidebar displays a list of files and directories. The file 'week36' is highlighted with a green circle, indicating its location within the repository structure.

File	Update date	Time ago
handin_1	Update date.	2 years ago
handin_2	Refine Option A instructions in README	11 months ago
handin_3	Update README.md	10 months ago
week35	Delete week35/DS_Store	last week
week36	Delete week36/Friday/DS_Store	last week
week37	Delete week37/Wednesday/week37.pdf	last week
week38	Delete week38/Monday directory	last week
week39	Update exercises.qmd	last year
week40	Update README.md	last year
week41	Add files via upload	11 months ago
week43	Update README.md	3 years ago
week44	Update README.md	11 months ago
week45	Update README.md	3 years ago
week46	Added paper question answers	3 years ago
week47	Update material.	2 years ago
week48	Add files via upload	10 months ago
week49/Wednesday	Create README.md	3 years ago
.DS_Store	updating the pp	3 years ago
.gitignore	Update material.	2 years ago
README.md	updating readme	last week



The screenshot shows the GitHub repository page for 'Evolutionary_Thinking_2026' at the path 'week36 / Wednesday'. The file 'Pretest.pdf' is highlighted with a green circle, indicating its location within the repository structure.

Evolutionary-Thinking-2026 -- Week 35

[Slides for today](#)

We will go through the pretest of "tree", basically a general understanding of the phylogeny. Then we will finish the installation of MEGA 11 for Friday's exercises.

Tree Pretest

Pen and Paper exercises

30 min group works with your peers before starting checking the answers.

HINTS:

1. Meaning of the nodes and branches in a phylogenetic tree.
2. Concept of "Common Ancestor"

[Solutions](#) to the pretest

MEGA 11 installation

Follow the guide from [MEGA](#) to download the corresponding version for your system.

There are also simple visual tutorials available for MEGA [download](#) and for [Windows installation](#).



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THE PLAN

			LECTURE	TA on lecture readings	Paper discussions	Hand-in worktime	Computer exercises	
Reading_Assignments								
#	Course week	#	Week number	Lecture Tuesday	Readings	TA Wednesday	TA Friday	Hand-Ins
	1		35	CANCELLED	Yang, pp 1-23 and origino life paper	Lecture on origin of life and substitution models Formation of study groups	Lecture Phylogenetic tree reconstruction	
	2		36	Lecture Trees continued plus Molecular clocks	Yang pp 23-50,	Introductory exercises	Tree pretest exercise (pen and paper) MEGA11 exercises: retrieve sequences, alignment and distance matrix	
	3		37	Paper discussion: Modeltest paper, Yang problems	Graur 165-209	MEGA11 exercises: substitution models	Work on hand-in 1	Hand-in I: Phylogenetic tree analysis
	4		38	Molecular clocks and gene duplication	Yang 361-368	Paper discussion: Molecular Epidemiology of HIV Transmission in a Dental Practice.	Introduction to R using phylogenetics	Hand-in I deadline
	5		39	Mikkel away: Basics of population genetics (lecture by Anders Charmouh Poulsen)	NS 2,3, Charlesworth paper	Problems NS 1,2,3 Hand-in I correction	Wright-Fisher and Coalescent simulator exercises (Python) + ADD questions to guide	
	6		40	Mikkel away: The coalescent continued (lecture by Anders Charmouh Poulsen)	NS 4,5,6, Ellegren paper	Problems NS 4,5,6	Problems NS 4,5,6	
	7		41	Tests of neutrality (lecture by Anders Charmouh Poulsen)	NS 7,8,9,	Problems NS 7,8	R exercises: detecting positive selection using dn/ds.	Hand-in II: NORK
-			42	Autumn Break		Autumn Break	Autumn Break	Autumn Break
	8		43	Incomplete lineage sorting	Mailund et al.	Discussion Mailund et al	Hand-in II discussion	Hand-in II deadline
	9		44	Comparative genomics and X chromosomes	Romiguiet et al. paper Vicoso paper	Discussion Romiguiet et al & Vicoso paper	Python exercises on population structure of goats	
	10		45	Primate genomics,	Juan et al and Kuderna et al	Discussion Juen et al., and Kuderna et al.,	PCA population structure in humans	
	11		46	Ancient DNA	Rasmussen et al.	Discussion Rasmussen et al.	R exercises SNPs	
	12		47			Paper discussion: Bergeron et al	Work on hand-in III	Hand-in III: Ancient DNA
	13		48	Genome evolution, TE	Barton chapter	Whatever is needed at this point/ make the selftest / any paper mikkel thinks is relevant	self-test discussion	Hand-in III deadline
	14		49	Evaluation/exam discussion and questions	none	Hand-in III correction/discussion, Summary of the exercises/discussion	FREE	

The github is updated up to here
The **exercises** and **solution** can be found there

We make the presentations during the week based on what you need more explanation about

We changed some things in the second part, the github is not updated for that yet!



TODAY PRETEST EXERCISES

See the github page

The answer sheet is also there but try to make it with your group first

About phylogentic trees:

1. Meaning of the nodes and branches in a phylogenic tree
2. Concept of "Common Ancestor"



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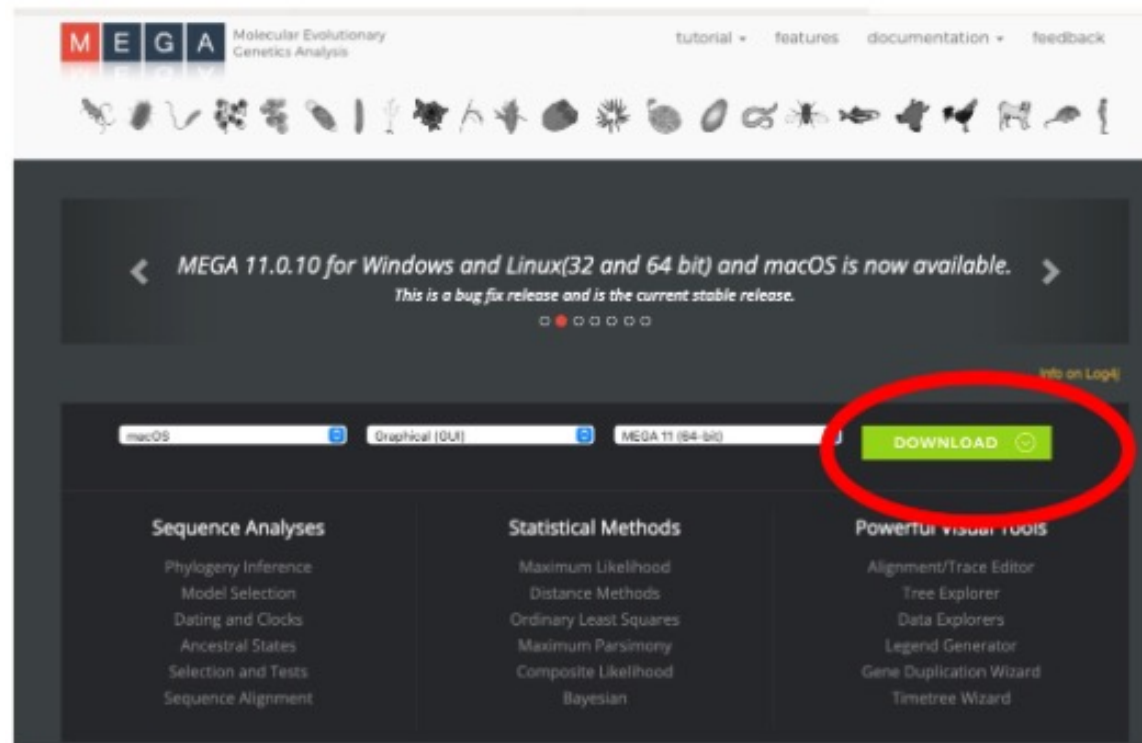
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DOWNLOAD MEGA

Google MEGA11 and download it

We will use this friday



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MENTI WITH QUESTIONS

What are things you want to have a bit more explanation about



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FRIDAY & NEXT WEEK

Start to practice with exercises in MEGA

Next week we focus on **DNA substitution models**



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